Web API Controller

Overview



- Web API Controller
- Web API Controller Characteristics
- Action Method Naming Conventions
- Request and Response pipeline
- Difference between Web API and MVC controller

Web API Controller

- In ASP.NET Core, a Web API controller is a class that handles incoming HTTP requests and produces HTTP responses. It serves as the main component responsible for processing client requests and generating appropriate responses in a Web API application
- An ASP.NET Core Web API controller typically derives from the **ControllerBase** class or the **ApiController** class, depending on the version of ASP.NET Core you are using.
- ASP.NET Core Web API controllers are a fundamental part of building RESTful APIs. They handle the routing and processing of incoming HTTP requests, interact with business logic or data access layers, and generate the corresponding HTTP responses.

```
[ApiController]
 public class EmployeeController : ControllerBase
 {
    [HttpGet]
    public IActionResult GetAllEmp()
          {
                //your logic goes here
          }
    [HttpGet]
          public IActionResult GetEmpById(int empCode)
          {
                //your logic goes here
          }
    [HttpPost]
          public IActionResult PostEmp([FromBody] EmpProfile empProfile)
          {
                //your logic goes here
          }
          [HttpPut]
          public IActionResult PutEmp([FromBody] EmpProfile empProfile)
          {
                //your logic goes here
         }
          [HttpDelete]
          public IActionResult DeleteEmp(int id)
          {
                //your logic goes here
          }
}
```

Web API Controller

Web API Controller Characteristics

- It must be derived from Microsoft.AspNetCore.Mvc.**ControllerBase** class.

- It can be created under any folder in the project's root folder.

- Action method name can be the same as HTTP verb name or
  it can start with HTTP verb with any suffix (case in-sensitive) or you can apply Http verb attributes to method.

- Return type of an action method can be any primitive or complex type.

Action Method Naming Conventions

HTTP Method	Possible Web API Action Method Name	Usage
GET	Get() get() GET() GetAllStudent() *any name starting with Get *	Retrieves data.
POST	Post() post() POST() PostNewStudent() *any name starting with Post*	Inserts new record.
PUT	Put() put() PUT() PutStudent() *any name starting with Put*	Updates existing record.

PATCH	Patch() patch() PATCH() PatchStudent() *any name starting with Patch*	Updates record partially.
DELETE	Delete() delete() DELETE() DeleteStudent() *any name starting with Delete*	Deletes record.

Differences between Web API and MVC controller

Web API Controller	MVC Controller
Derives from Microsoft.AspNetCore.Mvc.**ControllerBase** class	Derives from Microsoft.AspNetCore.Mvc.**Controller** class.
Method name must start with Http verbs otherwise apply http verbs attribute.	Must apply appropriate Http verbs attribute.
Specialized in returning data.	**Specialized in rendering view.**
Return data automatically formatted based on Accept-Type header attribute. Default to json or xml.	Returns ActionResult or any derived type.

References

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